

Reflection Journal

Hotel Booking Demand Classification Analysis

Weaving through the Data Landscape:

The adventure started with a dataset that on the one hand was misleadingly clean but on the other hand contained numerous misleading traps. The first significant challenge was the discovery of Data Leakage. When examining the features, I found that the reservation status column was a cheat code--it would record what had already happened. Adding it allowed the models to have a fake 100% accuracy rate that in a real-life scenario, gives zero predictive value. To eliminate this, I needed to change my thinking: I needed to use here such leading indicators as lead_time, deposit type, and total of special requests. This made me understand that the most potent feature in data science is often the feature that one should not use.

The Model Marathon:

The six different models that were trained were a masterclass of the No Free Lunch theory. I noticed a definite performance hierarchy. Logistic Regression was a good starting point, but it was not able to cope with the non-linear complexities of traveler behavior. K-Nearest Neighbors (KNN) and Support Vector Classifiers (SVC) were computationally intensive and did not scale well with the size of the data and needed careful normalization.

The Tree-based models (Random Forest and Gradient Boosting) were however the obvious contenders. Gradient Boosting, especially, demonstrated an amazing skill of reducing the errors by focusing on the hard to predict cases. This section of the project reaffirmed the rationale behind why ensemble methods are the standard approach to tabular data; it is resistant to outliers and it also captures interactions between variables that linear models cannot.

The Rationale of the Metrics:

One of the critical points in this project was the choice of the metric to focus on. Although Accuracy is the simplest to communicate to a stakeholder, it is perilous in hospitality. When we guess that a guest will not cancel when in fact they do, we lose an opportunity to resell that room (a False Negative). By changing the model to focus on Recall and the F1-Score, I made sure that the model was not overly aggressive.

Concluding:

The last phase that the workflow underwent was the final implementation of the Bayesian Stacking Ensemble. I did not simply allow the models to vote equally, but used a Naive Bayes meta-learner to weight the opinions of the models. Of particular interest was the way the ensemble balanced out the personal shortcomings of the base models. This was not a simple writing-code project; it was a project of creating a reliable decision-support tool that is balanced between statistical rigour and business reality.